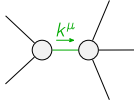
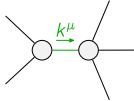
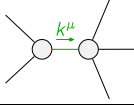
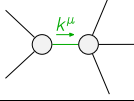
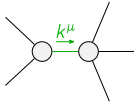
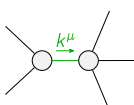
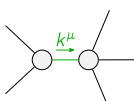


	$\mathcal{K}_0^{\#1}$	$\mathcal{K}_0^{\#1} \alpha\beta_X$				
$\mathcal{K}_0^{\#1} \dagger$	0	0				
$\mathcal{K}_0^{\#1} \dagger^{\alpha\beta_X}$	0	0	$\mathcal{K}_{1+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1-}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1-}^{\#2}{}_{\alpha}$
	$\mathcal{K}_{1+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{Y_1 k^2}{4}$	$-\frac{Y_2 k^2}{2\sqrt{2}}$	0	0	
	$\mathcal{K}_{1+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{Y_2 k^2}{2\sqrt{2}}$	$-\frac{Y_3 k^2}{4}$	0	0	
	$\mathcal{K}_{1+}^{\#1} \dagger^\alpha$	0	0	0	0	
	$\mathcal{K}_{1+}^{\#2} \dagger^\alpha$	0	0	0	0	
					$\mathcal{K}_{2+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2+}^{\#1} \alpha\beta_X$
				$\mathcal{K}_{2+}^{\#1} \dagger^{\alpha\beta}$	$\frac{3 Y_4 k^2}{4}$	0
				$\mathcal{K}_{2-}^{\#1} \dagger^{\alpha\beta_X}$	0	$\frac{3 k^2 \kappa_{15}}{2}^{(4)}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1} \alpha\beta_X$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1} \dagger^{\alpha\beta_X}$	0	0	$\mathcal{T}_{1^+}^{\#1} \alpha\beta$	$\mathcal{T}_{1^+}^{\#2} \alpha\beta$	$\mathcal{T}_{1^+}^{\#1} \alpha$	$\mathcal{T}_{1^+}^{\#2} \alpha$
	$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_3 k^2}{4 \text{Det}(1^+)}$	$\frac{\Upsilon_2 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{\Upsilon_1 k^2}{4 \text{Det}(1^+)}$	0	0	
	$\mathcal{T}_{1^+}^{\#1} \dagger^\alpha$	0	0	0	0	
	$\mathcal{T}_{1^+}^{\#2} \dagger^\alpha$	0	0	0	0	
					$\mathcal{T}_{2^+}^{\#1} \alpha\beta$	$\mathcal{T}_{2^+}^{\#1} \alpha\beta_X$
				$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
				$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta_X}$	0	$\frac{2}{3 k^2 \kappa_{15}^{(4)}}$

Abbreviations used in matrices	
$Y_1 = 6 \binom{(4)}{K_1} + \binom{(4)}{K_{15}} + 4 \binom{(4)}{K_3} \&\& Y_2 = 2 \binom{(4)}{K_{15}} - \binom{(4)}{K_4} \&\& Y_3 = 5 \binom{(4)}{K_{15}} - 2 \left(\binom{(4)}{K_3} + \binom{(4)}{K_4} \right) \&\& Y_4 = 2 \binom{(4)}{K_1} + 3 \binom{(4)}{K_{15}} \&\&$	
$\text{Det}(1^+) = \frac{1}{16} k^4 (-3 \binom{(4)}{K_{15}}^2 - 2 \left(2 \binom{(4)}{K_3} + \binom{(4)}{K_4} \right)^2 + 6 \binom{(4)}{K_{15}} (3 \binom{(4)}{K_3} + \binom{(4)}{K_4}) + 6 \binom{(4)}{K_1} (5 \binom{(4)}{K_{15}} - 2 \left(\binom{(4)}{K_3} + \binom{(4)}{K_4} \right))) \&\&$	
$\text{Det}(2^+) = \frac{3}{4} k^2 (2 \binom{(4)}{K_1} + 3 \binom{(4)}{K_{15}})$	

$$\begin{aligned} & \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^X \mathcal{K}^\alpha_{\alpha\beta} + \frac{3}{2} \overset{(4)}{\kappa}_{15} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^X \mathcal{K}^\alpha_{\alpha\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \\ & \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 3 \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{3}{2} \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \\ & 3 \overset{(4)}{\kappa}_{15} \partial^X \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \frac{9}{4} \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\ & \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{T}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{T}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{T}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{T}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{T}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{T}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{T}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{T}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{T}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{T}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{T}^{\chi\alpha\beta}$	
$\mathcal{T}_0^{\#1} == 0$	1	$\partial_\beta \mathcal{T}^\alpha{}_\alpha{}^\beta == 0$	
$\mathcal{T}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi} == 0$	
$\mathcal{T}_1^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{T}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{T}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi}$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{2 \binom{(4)}{\kappa_1} + 5 \binom{(4)}{\kappa_{15}}}{2 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2}$
	1	0	$-\frac{2 \binom{(4)}{\kappa_1} + 5 \binom{(4)}{\kappa_{15}}}{2 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2}$
	1	0	$\frac{\sqrt{4 \binom{(4)}{\kappa_1}^2 + 20 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 33 \binom{(4)}{\kappa_{15}}^2}}{2 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2}$
	1	0	$-\frac{\sqrt{4 \binom{(4)}{\kappa_1}^2 + 20 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 33 \binom{(4)}{\kappa_{15}}^2}}{2 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2}$
	1	0	$(96 \binom{(4)}{\kappa_1}^2 \binom{(4)}{\kappa_{15}} + 192 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}}^2 + 72 \binom{(4)}{\kappa_{15}}^3 + 64 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 96 \binom{(4)}{\kappa_{15}}^2 \binom{(4)}{\kappa_3} - 16 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_4} -$ $24 \binom{(4)}{\kappa_{15}}^2 \binom{(4)}{\kappa_4} - \sqrt{((2 \binom{(4)}{\kappa_1} + 3 \binom{(4)}{\kappa_{15}})^2 (1233 \binom{(4)}{\kappa_{15}}^4 + 12 \binom{(4)}{\kappa_{15}}^3 (199 \binom{(4)}{\kappa_3} - 79 \binom{(4)}{\kappa_4}) + 4 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 -$ $8 \binom{(4)}{\kappa_{15}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 (19 \binom{(4)}{\kappa_3} + 5 \binom{(4)}{\kappa_4}) + 12 \binom{(4)}{\kappa_{15}}^2 (303 \binom{(4)}{\kappa_3}^2 - 10 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 22 \binom{(4)}{\kappa_4}^2) +$ $36 \binom{(4)}{\kappa_1}^2 (209 \binom{(4)}{\kappa_{15}}^2 - 36 \binom{(4)}{\kappa_{15}} (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 4 (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2) + 12 \binom{(4)}{\kappa_1} (303 \binom{(4)}{\kappa_{15}}^3 +$ $4 \binom{(4)}{\kappa_{15}}^2 (218 \binom{(4)}{\kappa_3} - 5 \binom{(4)}{\kappa_4}) + 4 (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 2 \binom{(4)}{\kappa_{15}} (74 \binom{(4)}{\kappa_3}^2 + 84 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 19 \binom{(4)}{\kappa_4}^2))))) /$ $(\binom{(4)}{\kappa_{15}} (2 \binom{(4)}{\kappa_1} + 3 \binom{(4)}{\kappa_{15}}) (-30 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2 + 12 \binom{(4)}{\kappa_1} (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 2 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 6 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})))$
	1	0	$(96 \binom{(4)}{\kappa_1}^2 \binom{(4)}{\kappa_{15}} + 192 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}}^2 + 72 \binom{(4)}{\kappa_{15}}^3 + 64 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 96 \binom{(4)}{\kappa_{15}}^2 \binom{(4)}{\kappa_3} - 16 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_4} -$ $24 \binom{(4)}{\kappa_{15}}^2 \binom{(4)}{\kappa_4} + \sqrt{((2 \binom{(4)}{\kappa_1} + 3 \binom{(4)}{\kappa_{15}})^2 (1233 \binom{(4)}{\kappa_{15}}^4 + 12 \binom{(4)}{\kappa_{15}}^3 (199 \binom{(4)}{\kappa_3} - 79 \binom{(4)}{\kappa_4}) + 4 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 -$ $8 \binom{(4)}{\kappa_{15}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 (19 \binom{(4)}{\kappa_3} + 5 \binom{(4)}{\kappa_4}) + 12 \binom{(4)}{\kappa_{15}}^2 (303 \binom{(4)}{\kappa_3}^2 - 10 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 22 \binom{(4)}{\kappa_4}^2) +$ $36 \binom{(4)}{\kappa_1}^2 (209 \binom{(4)}{\kappa_{15}}^2 - 36 \binom{(4)}{\kappa_{15}} (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 4 (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2) + 12 \binom{(4)}{\kappa_1} (303 \binom{(4)}{\kappa_{15}}^3 +$ $4 \binom{(4)}{\kappa_{15}}^2 (218 \binom{(4)}{\kappa_3} - 5 \binom{(4)}{\kappa_4}) + 4 (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 2 \binom{(4)}{\kappa_{15}} (74 \binom{(4)}{\kappa_3}^2 + 84 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 19 \binom{(4)}{\kappa_4}^2))))) /$ $(\binom{(4)}{\kappa_{15}} (2 \binom{(4)}{\kappa_1} + 3 \binom{(4)}{\kappa_{15}}) (-30 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_{15}}^2 + 12 \binom{(4)}{\kappa_1} (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 2 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 6 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})))$
	2	0	$(18 \binom{(4)}{\kappa_{15}}$